

YOUNGYOON CHOI

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RESEARCH INTERESTS

Learning the world from visual experience

- **Generative Modeling** – Diffusion & flow models, controllable generation, representation learning
- **Visual World Models** – Toward long-horizon, world-consistent, and interactive generation

PAPERS

In-Distribution Forcing for Long Video Generation at Test-Time

Jeongwoo Shin*, **Youngyoon Choi***, Sangwoo Jo, Hyunmog Kim, Sungjoon Choi, Joonseok Lee, Jaewoong Choi, Jaemoo Choi

arxiv, 2026

Keywords: Long Video Generation, Exposure Bias, Test-time Extrapolation

Diverse Motion Customization via Control-based Dynamic Optimization

Youngyoon Choi*, Kihyun Kim*, Jeongwoo Shin, Joonseok Lee

arxiv, 2026

Keywords: Motion Customization, Stochastic Optimal Control, Controllable video generation

Improving 3D MRI Generation via Latent Space Decorrelation

Wonyoung Jang*, Junho Lee*, **Youngyoon Choi**, Seunggeun Lee, Joonseok Lee

Medical Image Understanding and Analysis (MIUA) 2026, **Oral**

Keywords: Brain Imaging, Flow matching, Representation Learning

GaussianVideo: Efficient Video Representation and Compression by Gaussian Splatting

Inseo Lee, **Youngyoon Choi**, Joonseok Lee

CVPR Workshop on Perception Beyond the Visible Spectrum 2025

Keywords: 3D Gaussian Splatting, Video Compression, Neural Rendering

EDUCATION

M.S. in Graduate School of Data Science

Mar 2024 – Present

Seoul National University, South Korea

Advisor: Joonseok Lee · Expected Feb 2027

B.A. in Psychology (Interdisciplinary Program in Brain, Mind, and Behavior)

Mar 2019 – Feb 2024

Seoul National University, South Korea

CGPA: 3.96 / 4.3 · Degree Honors: Summa Cum Laude

RESEARCH EXPERIENCE

Graduate Researcher, Visual Information Processing (VIP) Lab

Feb 2024 – Present

Advisor: Joonseok Lee

- Research on generative modeling for video, with a focus on diffusion and flow-based models.
- Developed a training-free framework to mitigate drifting in long-horizon video generation, extending short-horizon models to minute-scale generation.
- Developed a learning framework for motion customization that is structurally robust to content leakage, via stochastic optimal control.

Undergraduate Researcher, Computational Clinical Science (CCS) Lab

Sep 2023 – Dec 2023

Advisor: Wooyoung Ahn

- Conducted research on multimodal decision-making prediction, applying probabilistic modeling to integrate heterogeneous data.

- Received 3rd place at a research symposium.

Undergraduate Researcher, Human Factors Psychology (HFP) Lab

Jan 2023 – May 2023

Advisor: Sowon Hahn

- Assisted in participant recruitment and experimental procedures for a study examining how feedback modality in human-robot interaction affects exercise behavior change.

Undergraduate Researcher, Connectome Lab

Jan 2022 – Dec 2022

Advisor: Jiwook Cha

- Conducted research on youth suicide, applying machine learning to identify risk factors.
- Selected for the Undergraduate Research Funding Program.

SELECTED PROJECTS

LexiMate: An End-to-End Inclusive System for Dyslexia Support

Developer / Data Engineer / Domain Expert

- Built a product-level end-to-end system for dyslexia screening and intervention using real-time multimodal data.
- Designed and developed the data collection pipeline, enabling structured acquisition of heterogeneous multimodal inputs.

Efficient Motion Generation with Mean Flows

Team Researcher

- Developed a one-step text-conditioned 3D human motion generation framework using mean flow modeling, achieving real-time synthesis with reduced inference cost over diffusion-based methods.
- Collaborated on designing the continuous-time vector field model in latent motion space.

Advancing Image Reconstruction: Evaluating BOLD and Beta Signals in fMRI-Based Brain Decoding

Team Researcher

- Explored diffusion-based generative approaches for fMRI-to-image reconstruction, evaluating output quality via CLIP, SSIM, and semantic similarity.
- Designed and ran the multi-metric evaluation pipeline to quantitatively compare BOLD signal- and beta value-based reconstructions.

TEACHING EXPERIENCE

Basic Computing: First Adventures in Computing

Feb 2024 – June 2026

Led weekly hands-on Python programming classes for undergraduate students

Computational Thinking: Thinking with Computers

Dec 2025

Delivered a special lecture on data analysis and visualization

Machine Learning and Deep Learning I

Sep 2024 – Nov 2024

SCHOLARSHIPS & AWARDS

4th stage BK21 Scholarship

Sep 2024 – Dec 2025

3rd Place, Brain-Mind-Behavior Interdisciplinary Research Presentation

Dec 2023

Undergraduate Research Funding Program

Jul 2022 – Dec 2022

Dean's List, Academic Scholarship

Dec 2020

SKILLS

Languages	Korean (Native), English (Fluent), Spanish (Functional)
Programming	Python, R, SQL, C/C++